WEEK 3 PROGRESS REPORT

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

Group M Evening Week Ending: 19th September 2026

Work Completed Against Weekly Objectives

Task	Status	Notes
Corpus/Source Register	Completed	Eight public Makerere freshman onboarding documents were identified, collected, and registered with provenance metadata (title, source URL, purpose). Seven extracted successfully into plain text; the eighth (dynamic timetable portal) was excluded and noted in the register as un-embeddable static text.
RAG architecture diagram	Completed	Produced and committed. Shows the two parallel flows — offline document ingestion (download, extract, chunk, embed, index) and online query handling (embed question, similarity search, context builder, grounded generation, grounding check, answer or escalation).
Working RAG pipeline with source grounding	Completed	Implemented end-to-end across five scripts (download, extract, chunk, embed, retrieve) and an evaluation runner. Retrieval uses cosine similarity over Gemini embeddings with top-3 chunks above a 0.70 threshold. The grounded-answer prompt forces the model to answer only from retrieved context, cite source titles, and admit missing context rather than guessing.
15-case RAG evaluation	Completed	Completed with 5 answerable, 5 partially answerable, and 5 unanswerable questions. Results: 4/5 answerable passed, 2/5 partially answerable passed, 5/5 unanswerable correctly refused. Overall 11/15 (73%).
Failure catalogue	Completed	Three genuine failures documented with root cause analysis and proposed fixes (false refusal on an answerable question, corpus gap on password recovery, and exclusion of dynamic content).

Key Engineering Decisions

- Chose a conservative retrieval threshold (0.70) over a permissive one. After comparing 0.50, 0.65, and 0.70 against the corpus, we kept the strictest setting. For a support agent,

answering the wrong topic is worse than saying "I don't know" too often. The evaluation confirms this choice: zero unanswerable questions slipped through.

- Chunked by paragraph grouping to roughly 1500 characters rather than fixed-size token windows. This keeps each chunk coherent and self-contained, which matters because a chunk that starts mid-sentence cannot be cited meaningfully.

- Paired retrieval task types correctly (RETRIEVAL_DOCUMENT for chunks, RETRIEVAL_QUERY for questions) using gemini-embedding-001. This pairing is the intended use of the model and materially improves retrieval quality over using a single task type for both.

- Saved embeddings incrementally after each batch of ten chunks. When the free-tier rate limit was hit mid-run (429 RESOURCE_EXHAUSTED at chunk 71 of 90), the pipeline resumed from where it stopped the next day instead of losing all progress.

Failures, Challenges and Current Response

- False refusal on an answerable question ("What does a fresher need to bring during joining?"). The correct chunk exists in the joining instructions document but scored below 0.70 because the query phrasing did not overlap strongly with the chunk's formal wording. Response: documented in the failure catalogue with proposed fixes (increase top-k to 5, lower threshold to 0.68, or add metadata filtering). Fix scheduled for Week 4–5.

- Corpus gap on password recovery. The corpus contains the webmail activation guide but no recovery guide. Response: documented as a genuine gap, not a system bug. Proposed fix is either to add an ICT recovery guide if publicly available or to add a fallback escalation chunk.

- Dynamic content excluded from the corpus. Registration deadlines are published on a live university portal, which cannot be embedded as static text. Response: accepted as a scope limitation; deadline questions will be routed to a support ticket rather than answered.

- Free-tier quota exhaustion during embedding. The 90-chunk embedding run hit a 429 mid-way. Response: the incremental save design meant the pipeline resumed cleanly the following day without losing the 70 chunks already computed.

Links and Evidence

GitHub repository (branch: rag-evaluation):
https://github.com/SumayyaN/BSE41014_capstone_group_m_ee

Relevant files: src/rag/download_sources.py, src/rag/extract_text.py, src/rag/chunk_text.py, src/rag/embed_chunks.py, src/rag/retrieve.py, src/rag/run_eval.py, src/tests/rag_eval.md, docs/rag_failures.md.

ClickUp board (Team space, Week 3): https://app.clickup.com/1200410000000374/v1/6-1200410000023701-1

Individual Contribution Summary

Member	Task Owned	Evidence
Namata Sumayya	Corpus assembly and Source Register	https://github.com/SumayyaN/BSE41014_capstone_group_m_ee/blob/main/knowledge/source_register.md
Aryampa Joshua	RAG architecture diagram	https://github.com/SumayyaN/BSE41014_capstone_group_m_ee/blob/main/docs/rag_architecture.png
Nakaye Hildah	RAG pipeline (download, extract, chunk, embed, retrieve)	https://github.com/SumayyaN/BSE41014_capstone_group_m_ee/blob/main/src/rag/retrieve.py
Mushabe Moses	15-case evaluation, failure catalogue, and progress report	https://github.com/SumayyaN/BSE41014_capstone_group_m_ee/blob/main/src/tests/rag_eval.md

Plan for Next Week (Week 4 - Tools and Function Calling)

- Define at least two tools with purpose, input schema, output schema, authorization, and failure behaviour.
- Implement tool/function calling through the application's orchestration layer.
- Ensure one tool performs a low-risk simulated side effect (ticket creation) and the other retrieves current application data.
- Add human approval before any higher-impact action, per the AI Boundary Matrix.
- Update the architecture diagram to reflect the tool layer and set up the Week 4 ClickUp board.